

2.1 Elements of computational thinking

Understand what is meant by computational thinking

2.1.1 Thinking abstractly	a) The nature of abstraction. b) The need for abstraction. c) The differences between an abstraction and reality. d) Devise an abstract model for a variety of situations.
2.1.2 Thinking ahead	a) Identify the inputs and outputs for a given situation. b) Determine the preconditions for devising a solution to a problem. c) The nature, benefits and drawbacks of caching. d) The need for reusable program components.
2.1.3 Thinking procedurally	a) Identify the components of a problem. b) Identify the components of a solution to a problem. c) Determine the order of the steps needed to solve a problem. d) Identify sub-procedures necessary to solve a problem.
2.1.4 Thinking logically	a) Identify the points in a solution where a decision has to be taken. b) Determine the logical conditions that affect the outcome of a decision. c) Determine how decisions affect flow through a program.
2.1.5 Thinking concurrently	a) Determine the parts of a problem that can be tackled at the same time. b) Outline the benefits and trade offs that might result from concurrent processing in a particular situation.